

Meridian Casualty: Hiring Is Broken. Can You Fix It?

Econ Week 2026 case competition, Round 1 written analysis. Team: Ethan Friedman, Austin Glick, Danny Shin and Sam Stark. Every number cites an appendix exhibit, which carries sample sizes, standard errors and confidence intervals.

Recommendation: make applicants bear it. Re-time the situational section, block pasting, and flag fast answers

Meridian should move the situational section (the essay, in the data) on its new applicant tracking system (ATS) back to the timed, no-paste format its seven legacy centers still use, and show recruiters the minutes each applicant spent in the section. That is option (b). Against the status quo it lowers the cost of one retained employee by **\$713 to \$1,556** when operational quality is left out, and by **\$2,548 to \$3,392** when it is monetized (E19). At Meridian's current new-ATS hiring pace that is **\$4.9 million to \$10.6 million a year** before quality and before the vendor's implementation charge, which the data do not contain (E19). The low end is the difference-in-differences gain alone; the high end is what the seven legacy centers achieve today (E19 rows b-low to b-high). The quality figures also rest on the legacy centers' lower reopen rate, whose difference-in-differences interaction on its own does not clear zero (E02b), so the excluding-quality range is the one we defend.

In one sentence: since 2023 the essay no longer predicts who stays on the new system, it still carries part of that signal on the legacy system, and the legacy centers retain more hires today (E12, E12b, E12d, E01b, E17). Every other option's retention effect would be an assumption; (b)'s is observed in Meridian's own centers.

The biggest hole: the seven legacy centers are small, sit in the Southeast and Mid-Atlantic, and their own essay signal is eroding. Matching them to new-ATS centers on size and region does not shrink their advantage, but seven centers can only bound it (E15f, E07b, E12e). Copying their configuration is therefore an upper bound, and we report the range rather than its midpoint.

Per retained employee, 2025 rates	Status quo	Option (b)	Saving
Excluding operational quality	\$23,377	\$21,821 to \$22,665	\$713 to \$1,556
Including operational quality	\$40,908	\$37,516 to \$38,360	\$2,548 to \$3,392
Annual, 6,830 retained seats			\$4.9M to \$10.6M (excl. quality)

Source: E17, E19. Six-month retention $r = 0.820$ at the status quo; 0.843 to 0.857 under (b).

(a) The process still measures the right things, but it weights a signal that no longer exists and reads it with half the time

What works. Four screen inputs still predict who lasts six months, and each predicts better in the AI era than before it. The one weak cell is the interview on the legacy system before 2023, where the slope is not distinguishable from zero (E11b). Holding the other inputs fixed, an adjuster license raises six-month retention by 1.6 points on the legacy system before 2023 and 4.9 points on the new system after 2023; a referral goes from 1.6 to 3.2 points, the resume score from 1.0 to 1.9 points per unit, and the structured interview from 0.9 to 2.7 points per unit (E11, E11b, E14). Volume is not the problem. Openings fill in about a month and each draws over a hundred applications (E10, E10b).

Where it fails, first judgment: the screen has not adapted. The recruiter screen score is a fixed formula. The essay's weight inside it is 0.250 in every ATS-and-era cell, with an R-squared of 0.82 to 0.86, exactly what it was when the essay worked (E11c). Meanwhile the essay's own predictive slope on retention has gone from 0.8 points per rubric point on the legacy system before 2023 to zero on the new system after 2023 (E11b). Meridian is still paying a quarter of its screen for information that is no longer there.

Second judgment: recruiter capacity is spread thinner every year, and thinnest where the essay is least informative. Applications per opening rose from 150 in 2021 to 217 in 2025 while recruiter minutes per application fell from 6.2 to 3.0 and days to fill rose from 22 to 30 (all centers pooled, E10e; by ATS, E10b). Each extra applicant per opening costs 0.023 minutes of review per application (E10f). More applicants and fewer minutes per application both lengthen the time to fill (E10c). The new ATS itself draws 348 more applications per center-month than the legacy system in the AI era, most likely because an untimed section that accepts pasted text is easier to finish (E10d). So the least informative essays arrive in the centers with the least time to read them.

The consequence is what Whitfield sees. Six-month retention on the new ATS fell 2.3 points more than on the legacy system once the AI era began (difference-in-differences -0.023 , 95% CI -0.042 to -0.004 , 41,677 hires, E01, E01b). Reopen rates on the new ATS are 0.14 to 0.25 percentage points higher in the AI era; the lower figure is the interaction, the upper the raw AI-era gap, and only the upper clears zero (E02, E02b). Separations rose across voluntary, performance and attendance exits, with no single reason carrying the increase (E03, E03b, E03c). That fits a screen that admits people who neither stick nor perform, not a labor-market story. Her "best-run centers are worst" is the pilot wave: the largest centers migrated first in 2021, held retention at 0.878 through 2022, then fell to 0.830 by 2024-25, while never-migrated centers moved from 0.889 to 0.857; the decline appears in every size tercile (E04, E05).

(b) The situational section no longer screens on the new system, and we know because its slope on outcomes collapsed in one cell and nowhere else

The comparison. We estimate the slope of the essay rubric score on the six-month outcome within each of four cells: legacy or new ATS at the application month, crossed with application before or after January 2023. **On whom.** Two samples. Permanent hires (41,677 with an observed outcome, after dropping 4,589 censored hires, E00) were selected on the essay, so every slope in that sample is attenuated. Flex placements (28,672) applied to Meridian but were placed by the staffing partner on certification, geography and availability, not on the essay, so their slope is the cleaner one. Both use center and month fixed effects with standard errors clustered by center.

The result. Among hires, a rubric point raised retention by 0.87 points on the legacy system before 2023 (95% CI 0.69 to 1.05) and 1.07 points on the new system before 2023 (0.74 to 1.40). On the new system after 2023 the slope is -0.26 points (-0.49 to -0.02): a higher essay score now predicts slightly worse retention (E12b), and the same reversal appears on hires' reopen rates (E12c). On the legacy system after 2023 it is 0.48 points (0.14 to 0.82), roughly half its old value (E12b). Flex placements tell the same story on reopen rates: each rubric point lowered the reopen rate by 0.18 points on legacy and 0.21 on the new system before 2023, by 0.12 on legacy after 2023, and by only 0.07 on the new system after 2023 (E12d). E12 shows the same thing without a regression. A 24-30 essay retains better than an 18-23 essay in three cells and worse in the fourth, 0.829 against 0.853. Combining the two samples, the new ATS in the AI era keeps **0% to 39%** of the essay's pre-2023 predictive power and the legacy ATS keeps **55% to 68%** (E15e). The two samples bracket the truth: hires understate the loss and flex placements do not.

The mechanism is time in the section. Among hires who spent 15 minutes or more writing, the essay still predicts retention in every cell, including the new system in the AI era (0.41 points per rubric point, 0.20 to 0.61). Among those who spent under 15 minutes on the new system after 2023, a higher score predicts worse retention (-1.64 points, -2.06 to -1.22) (E07, E13, E13b). Effort used to earn points. On the legacy system before 2023 a minute in the section added 0.054 rubric points; on the new system after 2023 a minute subtracts 0.021 (E13c). A fast high score is what a prepared answer pasted into the box looks like. Meridian already records the minutes and does not use them.

Placebo. If something center-wide had changed, every input would have lost power. None did: the resume slope went from 0.9 to 2.3 points, license from 2.0 to 6.1, prior claims years from 0.4 to 0.8, interview from -0.5 to 3.2, over the same cells in which the essay went from 0.87 to -0.26 (E14).

Timing. Essay scores rise from 2023 in every new-ATS center regardless of when it migrated, including the 2021 pilots, so the break is the AI era meeting an untimed field, not the software itself (E06). A grid search over break dates finds a ramp rather than a jump, so any cutoff in 2023 tells the same story (E06b). With center, month and applicant controls, the new ATS raised scores by 0.89 rubric points before 2023 and a further 0.79 (0.58 to 1.00) once AI writing arrived (E06c), while objective attributes drifted slightly down (license -2.6 points, prior claims experience -2.4, referral -1.1, E08, E08b) and the flex pool's reopen rate did not move (E09, E09b). The pool did not get better; the scores did.

Identification. Before 2023 the pilot centers' essay slope matched the never-migrated centers' (0.010 against 0.008 per rubric point, E15g), so migration did not select on essay validity. The event study shows no pre-trend, then a 0.029 widening after migration, which matches the difference-in-differences estimate from a different design (E15, E15b). Unemployment and wages move together across ATS groups (E16, E16b). The essay's standard deviation among hires is 2.85 to 2.98 in every cell, so the zero slope is not range restriction. What changed is who scores high: 23% of new-ATS AI-era applicants score 24-30, against 8% of legacy applicants before 2023 (E15c). With 40 centers, a wild-cluster bootstrap on the slope difference between the new-ATS AI-era cell and the legacy pre-2023 cell gives p below 0.01 in both samples (E15d).

What must be true for this to mean the essay stopped screening: that the staffing partner does not place on the essay (stated in the case); that nothing unobserved changed at migrating centers in 2023 that also changed essay validity (E15 and the pre-period parity above); and that the 2023 cutoff is substantively right (E06b). **What we could not determine:** whether any individual used AI; what the 93% to 95% of applications never reviewed look like (E10e); and how far the legacy system's own erosion continues, since its sub-10-minute share doubled from 6% to 12% and its bootstrap p on erosion is 0.08 for hires and 0.003 for flex (E07b, E15d). A timed field slows the leak; it does not stop it.

(c) Option (b) is the only choice whose retention effect is observed, and it wins under every bound

We implement Meridian's worksheet exactly and reproduce the worked example (E17a). The status quo is new-ATS hires starting in 2024-25: $r = 0.820$, 28.9 days to fill, an 8.9% reopen rate, and a cost per retained employee of \$23,377 excluding quality and \$40,908 including it (E17). Three counterfactuals in which the essay still screened put the cost of the lost information at \$713 to \$1,830 per retained employee excluding quality and \$2,548 to \$4,334 including it (E17). We show both because the quality term roughly doubles every gap and can change the ranking.

E19 costs the four options against the status quo per retained employee. **(a) Do nothing** is flat: retention did not move from 2024 to 2025 (+0.000), so extending that trend shifts the cost by \$23, which is noise. **(b) Applicants bear it** saves \$713 to \$1,556 excluding quality, from the difference-in-differences gain alone up to the legacy centers' 2024-25 retention, reopen rate and shorter fill time. The size-and-region matched estimate, \$1,379, sits inside that range (E15f, E19). **(c) Meridian bears it**, a structured interview at 15 extra recruiter minutes per reviewed application (\$204 per hire) and 3 more days to fill, costs \$448 more excluding quality and \$469 less including it, if it recovers half the gap; that recovery is assumed, not observed. **(d) Bear it after hiring**, a \$500 six-month bonus, costs \$305 more either way, because separations are front-loaded: 47% of leavers are gone by month two and only 33% leave in months four to six, before a probation review can act (E18, E18b). Recruiter screening, which the prompt puts inside C_hire , is about \$740 per hire (E17b). We leave it out of every scenario. It would fall under (b), which draws fewer applications per opening (E10b, E10d), so leaving it out understates (b)'s saving; (c) is charged its extra minutes.

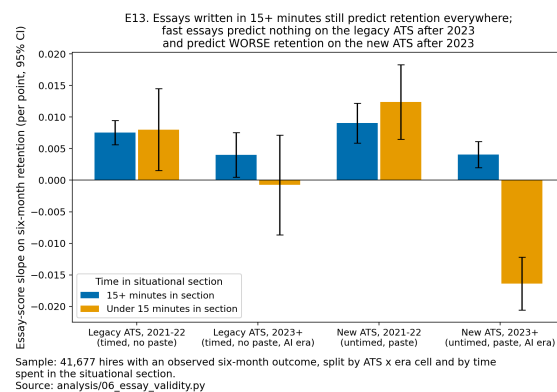
Sensitivity. The prompt's warning holds: (b) trades fewer separations for possibly more vacancy days. At \$219 per vacancy day, (b)'s saving is erased by 2.7 extra days to fill if it only buys the difference-in-differences gain, 4.4 days at the legacy-observed retention, 5.5 at the matched estimate, and 7.4 at the pre-AI level; including quality, 9.8 to 14.8 days (E20, E20b, E20c). In the data the legacy centers fill faster, not slower (27.3 against 28.9 days in the 2024-25 cohorts, E17), and the matched gap is still about 0.7 days in their favor (E15f). The "fewer applicants, longer fill" fear does not show up, though those centers are small.

What (b) costs applicants. A timed, no-paste section is harder to complete; Whitfield is right about that. The first cost is drop-off. The legacy centers drew 183 applications per opening against 224 on the new system in 2025 (E10b), so a timed field could cost about a fifth of the queue. That is already priced in: the retention, reopen and fill figures (b) borrows from the legacy centers (E19) are what they achieve with the smaller queue, and sourcing cost is per opening, not per applicant (E17a). The second is fairness. A timer penalizes slow writers, including applicants who need accommodation, so the vendor ask should include extended time on request and the pilot should report completion rates by applicant group. We could not measure either cost; drop-off is our fifth Round 2 trigger.

We would recommend something else if:

- **(c) instead**, if a timed section pushed days to fill up by more than the break-evens above, or if a pilot showed a structured interview recovers more than 89% of the legacy retention gap at \$204 per hire (E20c).
- **(d) instead**, if separations were concentrated in months four to six rather than the first two (E18).
- **Do nothing**, if the Round 2 packet showed the legacy centers' advantage disappears once matched on size and region (on our data it does not, E15f), or that the vendor cannot re-time the section for less than the annual saving in E19.

Implementation, in three steps. First, at no cost, show recruiters the minutes-in-section field Meridian already collects and flag answers under 15 minutes (E13b). Second, ask the vendor for a fixed quote to make the section timed and paste-disabled on the new ATS; the annual saving in E19 is the most that change can cost and still pay. Third, pilot the timed section for one hiring quarter in the centers that migrated in 2024, with two stop rules: days to fill rising by more than the break-even in E20c, or applications per opening falling below the legacy centers' level (E10b). The nine 2024-wave centers make about 425 hires a quarter (E00), enough to detect a five-point retention gain but not the 2.3-point lower bound, so use the first quarter to check the stop rules and read retention against the difference-in-differences bound over two to three quarters before rolling out. Five things in the Round 2 packet could change this call: the vendor's cost and feasibility, the matched comparison, a labor-market shock, a revised reopened-claim cost, and applicant drop-off. We have written down what each would do to it.



E13. Slow essays (15+ min) still predict retention in every cell; fast essays on the new ATS after 2023 predict worse retention (E13b).

Appendix: exhibits

Exhibits appear in register order (E00 to E20), which follows the narrative's sections. Every figure carries its title, axis units, sample statement and source script inside the image; every table is reproduced from outputs/tables/ exactly as the analysis scripts wrote it, with its sample, fixed effects, controls and standard-error type in the notes beneath. Regressions are linear probability models for binary outcomes; standard errors are clustered by center unless a note says otherwise.

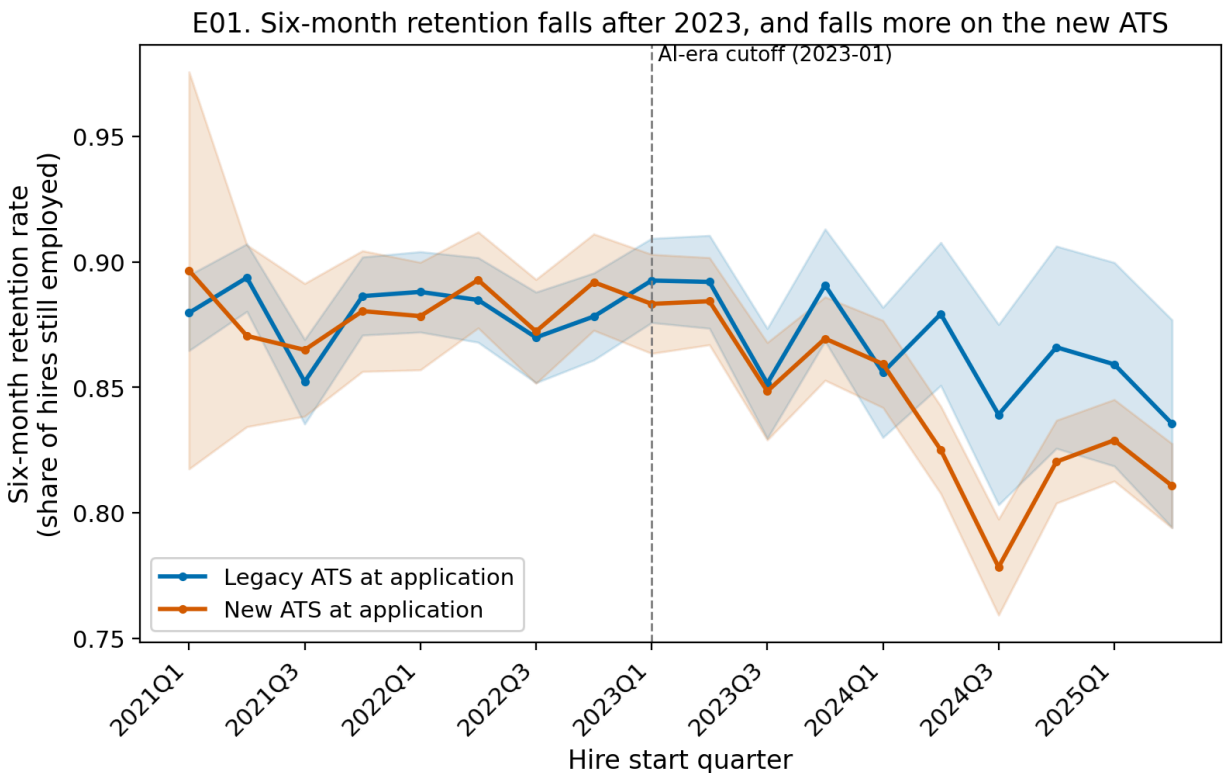
Exhibit	Title
E00	Sample construction and data-integrity checks
E01	Retention by quarter ATS (figure)
E01b	Retention DiD: new ATS x post-AI-era interaction
E02	Reopen by quarter ATS (figure)
E02b	Reopen-rate DiD: new ATS x post-AI-era interaction
E03	Separation reason by year (figure)
E03b	Separation reason share by start year (%)
E03c	Separation-reason DiD: new ATS x post-AI-era interaction, by reason
E04	Retention by wave year (figure)
E05	Retention by wave size tercile (figure)
E06	Essay score by ATS monthly (figure)
E06b	Grid search for the AI-era break date in new-ATS essay scores
E06c	Essay-score inflation: new ATS x post-AI-era interaction
E07	Essay minutes distribution (figure)
E07b	Mean essay score and share of essays under 10 minutes, by year and ATS
E08	Objective attributes over time (figure)
E08b	Objective applicant attributes: new ATS x post-AI-era DiD
E09	Flex reopen over time (figure)
E09b	Flex reopen rate: new ATS x post-AI-era interaction
E10	Recruiter capacity panel (figure)
E10b	Recruiter-capacity metrics by year and ATS
E10c	Drivers of days-to-fill
E10d	Applications received per center-month: new ATS x AI-era DiD
E10e	Recruiter-capacity metrics by year, all 40 centers pooled
E10f	Recruiter minutes per application vs applications per opening
E11	Screen component coefficients by regime (figure)
E11b	Predictors of six-month retention, by regime
E11c	How much weight recruiters place on the essay score, by regime
E12	Retention by essay bin regime (figure)
E12b	Essay-score slope on six-month retention, by regime (hires)
E12c	Essay-score slope on six-month reopen rate, by regime (hires)
E12d	Essay-score slope on reopen rate, by regime (flex placements -- NOT selected on the essay by Meridian)
E12e	Spec C: essay slope on retention in the 12 months before vs after a center's own ATS migration (2023/24 waves) vs never-migrated centers over the same calendar window
E13	Essay slope by speed regime (figure)
E13b	Essay-score slope on retention, by regime and by whether the applicant spent under 15 minutes in the situational section
E13c	Essay score per minute spent in the section, by regime
E14	Placebo test: slope of other screen inputs on six-month retention, by regime
E15	Migration event study (figure)
E15b	Event-study means: retention before/after migration vs never-migrated
E15c	Selection on the essay into the hire sample, by regime
E15d	Small-cluster inference: wild-cluster bootstrap on the headline essay-slope differences
E15e	Share of the legacy pre-2023 essay signal retained, by cell: hires sample vs flex sample
E15f	Matched legacy comparison (figure with table)
E15g	Pre-2023 essay slope on retention: pilot centers vs never-migrated centers
E16	Labor market by ATS (figure)
E16b	Labor-market conditions by year and ATS
E17	Cost per retained employee: status quo vs counterfactuals where the essay still screened
E17a	Worksheet inputs: 2025 average rates across 40 centers vs the prompt's worked example

Exhibit	Title
E17b	Recruiter screening time per hire (not in the prompt's C_hire; shown for option c)
E18	Separation hazard by month (figure)
E18b	Separations by month on the job and reason (counts)
E19	Cost per retained employee by option, vs status quo
E20	Sensitivity option b (figure)
E20b	Sensitivity table: cost per retained employee under option (b), \$ excl. quality, by r (rows) and extra days to fill (columns)
E20c	Break-even: extra vacancy days that erase option (b)'s saving; recovery share option (c) needs

E00. Sample construction and data-integrity checks

check	value	expected
applications.csv rows	529,302	529,302
hires.csv rows	46,266	46,266
flex_placements.csv rows	28,672	28,672
center_month.csv rows	2,400	2,400
ats_migration.csv rows	40	40
hires_joined rows (== hires.csv)	46,266	46,266
flex_joined rows (== flex_placements.csv)	28,672	28,672
applications with outcome=hired	46,266	46,266
applications with outcome=flex_placement	28,672	28,672
hires censored (no 6mo outcome)	4,589	expect ~4589
hires with observed 6mo outcome (analysis sample)	41,677	expect ~41677
coaching_hours null total	4,161	n/a
coaching_hours null AND censored	571	n/a
coaching_hours null AND NOT censored (true missing)	3,590	n/a
hires: ATS at start != ATS at application (use application-month ATS)	530	expect ~530
application-to-start lag distribution (months)	0 mo: 11,169; 1 mo: 31,475; 2 mo: 3,609; 3 mo: 13	n/a
hires_obs by regime (cutoff 2023-01)	legacy: 19,366; new_pre: 5,920; new_post: 16,391	n/a
flex by regime (cutoff 2023-01)	legacy: 14,620; new_pre: 2,346; new_post: 11,706	n/a
centers never migrated to new ATS	7	expect 7
hires starting in 2025 at the 9 centers that migrated in 2024 (all, incl. censored)	1,698	n/a
= hires per quarter at those centers	424	n/a

Sample: All five raw files, 2021-01 to 2025-12, 40 centers. *Notes:* Analysis sample for hire outcomes = hires_obs (retained_6mo observed); censored hires (started 2025-07 or later) are excluded from every hire regression and never imputed. *Source:* analysis/00_load.py

E01. Retention by quarter ATS

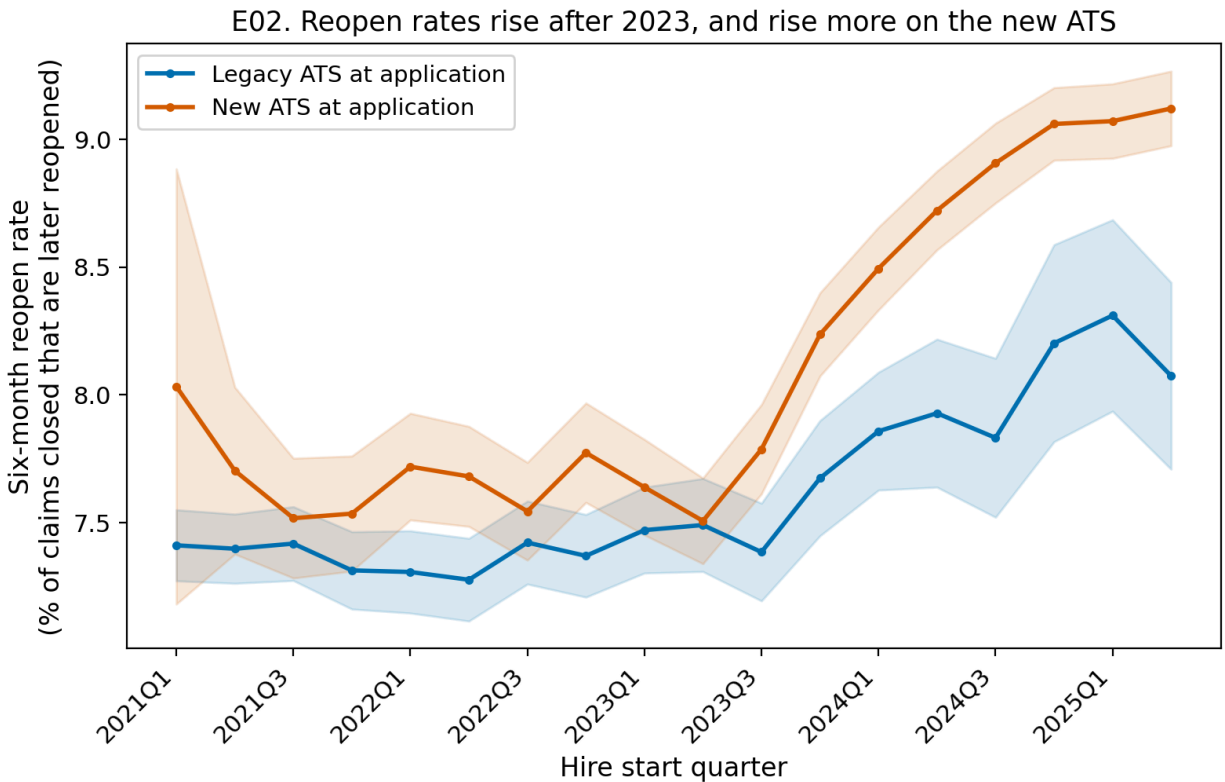
Sample: 41,677 permanent hires with an observed six-month outcome, start months 2021-01 to 2025-06 (later starts are censored).

Source: analysis/02_retention_trend.py

E01b. Retention DiD: new ATS x post-AI-era interaction

term	estimate	SE	CI low	CI high	p	n	clusters
new ATS effect, pre-AI era (2021-22)	0.003	0.009	-0.015	0.020	0.772	41,677	40
DiD interaction: extra new-ATS effect, AI era	-0.023	0.010	-0.042	-0.004	0.018	41,677	40
post-era gap, new vs legacy (sum of the two)	-0.021	0.009	-0.038	-0.004	0.017	41,677	40

Sample: n=41677, 40 center clusters. *Notes:* Outcome: retained_6mo (0/1, linear probability). FE: center, start_month; post_ai (application-month era) entered explicitly because start-month FE do not absorb it for hires who applied before and started after the cutoff. Controls: resume/license/experience/referral/employed/wage/unemployment. Cluster: center. Row 1 = new-ATS effect before the AI era; row 2 = the DiD interaction; row 3 = their sum, the new-vs-legacy gap in the AI era. *Source:* analysis/02_retention_trend.py

E02. Reopen by quarter ATS

Sample: 41,677 permanent hires with an observed six-month outcome.
Source: analysis/02_retention_trend.py

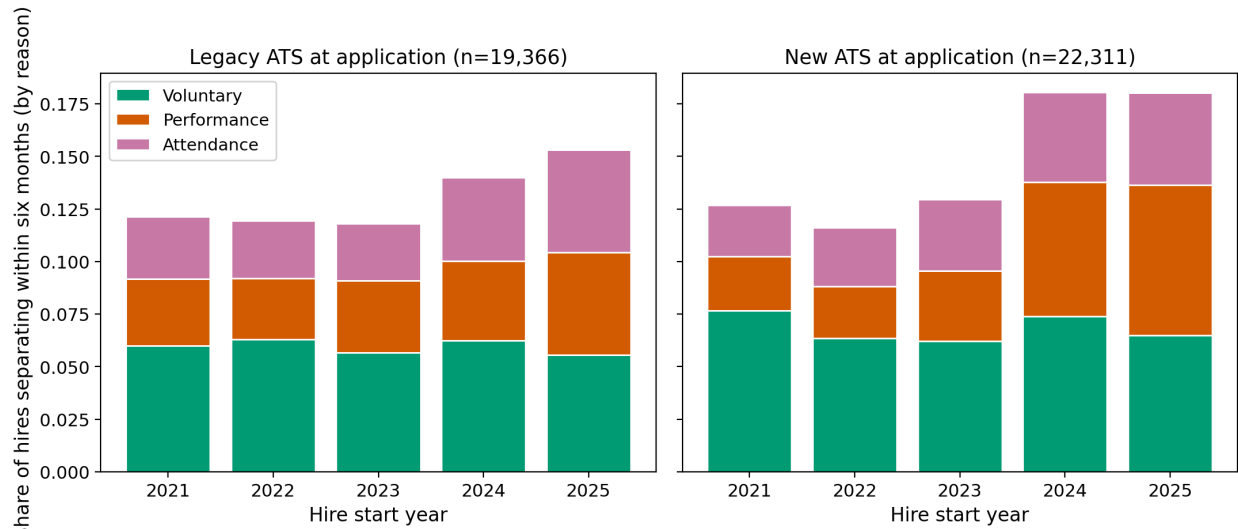
E02b. Reopen-rate DiD: new ATS x post-AI-era interaction

term	estimate	SE	CI low	CI high	p	n	clusters
new ATS effect, pre-AI era (2021-22)	0.103	0.079	-0.051	0.257	0.189	41,677	40
DiD interaction: extra new-ATS effect, AI era	0.145	0.091	-0.033	0.323	0.111	41,677	40
post-era gap, new vs legacy (sum of the two)	0.248	0.090	0.071	0.425	0.006	41,677	40

Sample: n=41677, 40 center clusters. *Notes:* Outcome: reopen_rate_6mo_pct (percentage points). Same FE/controls/cluster as E01b. *Source:* analysis/02_retention_trend.py

E03. Separation reason by year

E03. Separations rise after 2023 in every reason category, more on the new ATS



Sample: 41,677 permanent hires with an observed six-month outcome, by start year (2025 partial: only Jan-Jun starts have an outcome).

Source: analysis/02_retention_trend.py

E03b. Separation reason share by start year (%)

year	voluntary	performance	attendance
2021	6.32	3.08	2.85
2022	6.32	2.73	2.76
2023	5.97	3.39	3.08
2024	7.16	5.86	4.21
2025	6.36	6.89	4.43

Sample: 41,677 hires with an observed six-month outcome. Source: analysis/02_retention_trend.py

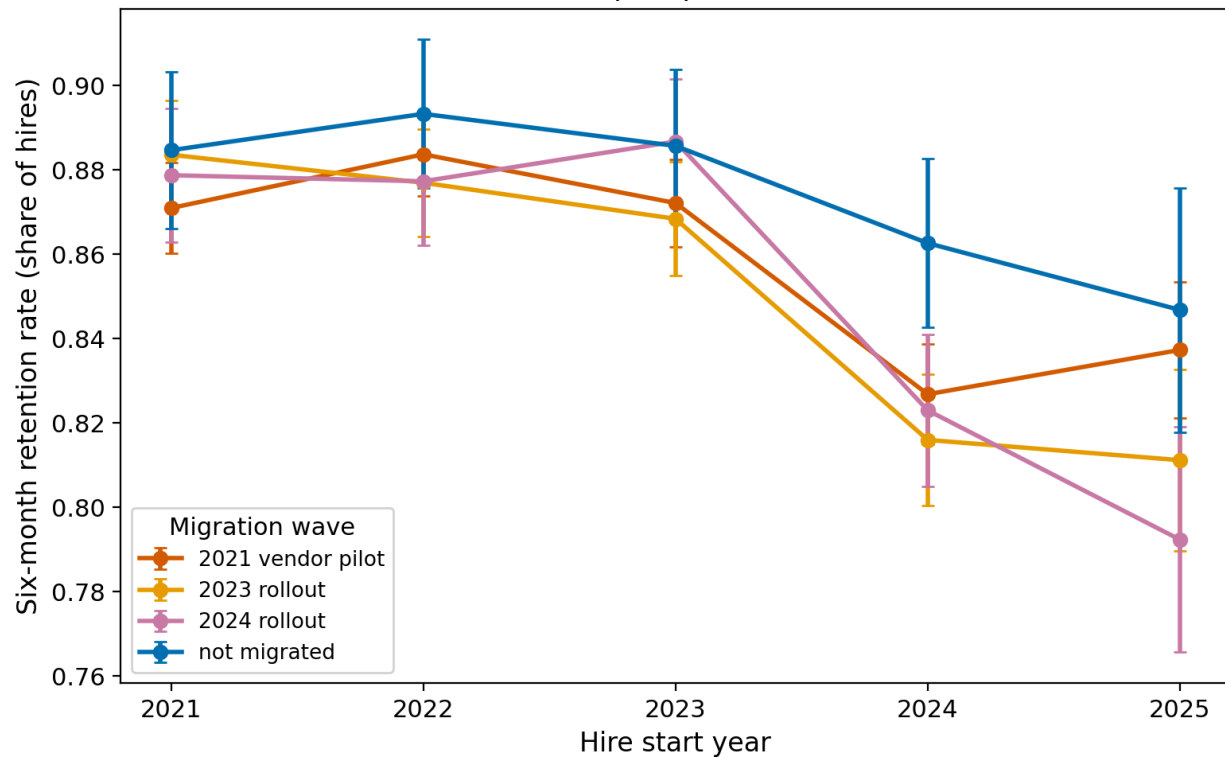
E03c. Separation-reason DiD: new ATS x post-AI-era interaction, by reason

	DiD estimate	CI low	CI high
voluntary	0.010	-0.000	0.020
performance	0.010	-0.001	0.021
attendance	0.003	-0.005	0.012

Sample: n=41677, 40 center clusters. Notes: Outcome: indicator for separating within six months for the stated reason. Same spec as E01b. Source: analysis/02_retention_trend.py

E04. Retention by wave year

E04. The 2021 pilot centers -- the largest, best-resourced centers -- show the steepest post-2023 decline

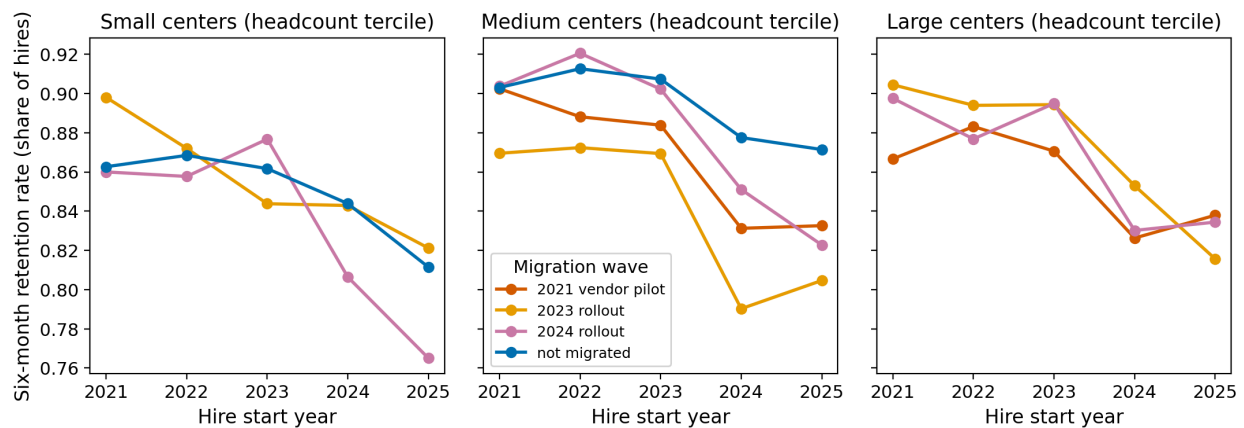


Sample: 41,677 permanent hires with an observed six-month outcome, grouped by their center's ATS migration wave.

Source: analysis/02_retention_trend.py

E05. Retention by wave size tercile

E05. The pilot-wave decline shows up within every size tercile -- size alone does not explain it

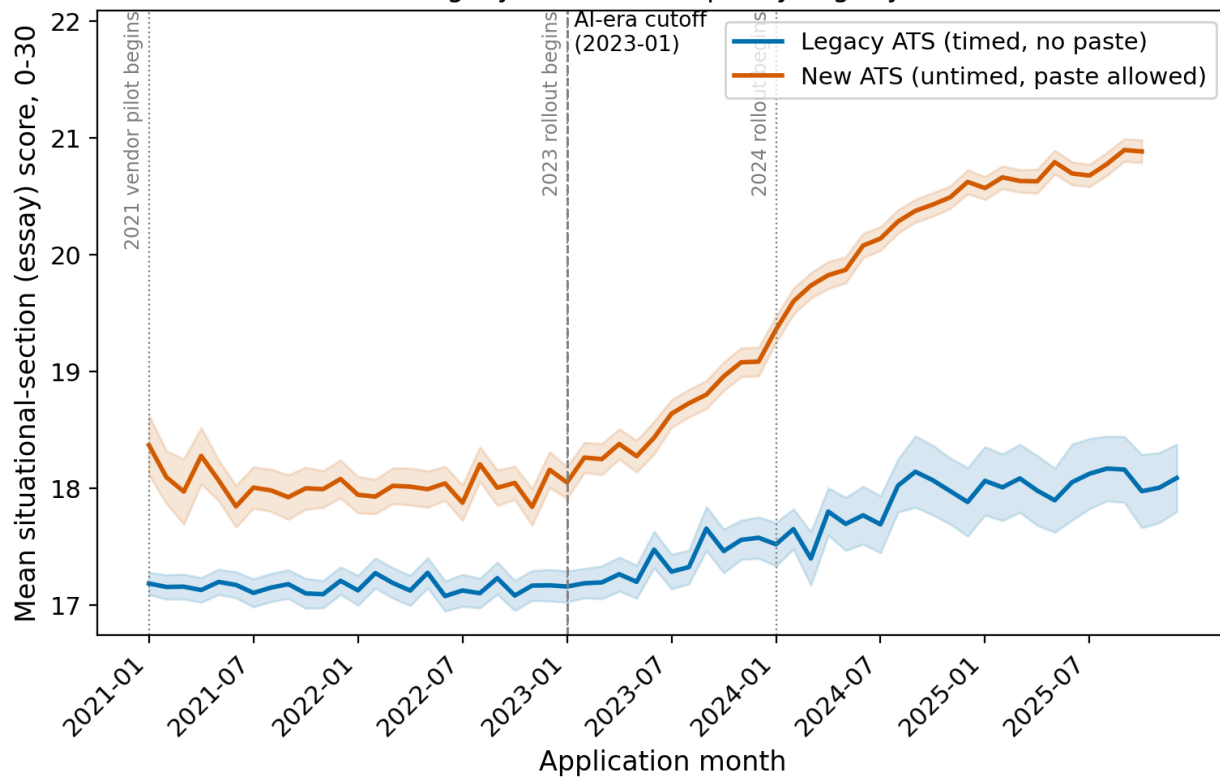


Sample: 41,677 permanent hires, split into center-headcount terciles.

Source: analysis/02_retention_trend.py

E06. Essay score by ATS monthly

E06. Essay scores climb on the new ATS from 2023;
legacy scores drift up only slightly



Sample: 529,302 reviewed applications, 2021-01 to 2025-12, 40 centers. Shaded band is a 95% CI of the monthly mean.

Source: analysis/03_pool_quality.py

E06b. Grid search for the AI-era break date in new-ATS essay scores

candidate month	pre mean	post mean	t
2022-07	18.01	19.73	69.81
2022-08	18.01	19.75	73.13
2022-09	18.01	19.77	76.58
2022-10	18	19.8	80.37
2022-11	18.02	19.82	83.03
2022-12	18.01	19.85	86.22
2023-01	18.02	19.88	89.63
2023-02	18.01	19.9	92.93
2023-03	18.02	19.92	95.48
2023-04	18.02	19.96	99.17
2023-05	18.03	19.99	102.3
2023-06	18.05	20.03	105.4
2023-07	18.07	20.07	108.2
2023-08	18.08	20.1	111.2
2023-09	18.09	20.14	113.6
2023-10	18.12	20.18	115.9
2023-11	18.16	20.23	117.9
2023-12	18.19	20.27	119.7

Sample: Applications reviewed under the new ATS, 2021-01 to 2025-12. Notes: Best-fit break month by max t-stat: 2023-12. Configured cutoff in analysis/utls.py is 2023-01. Source: analysis/03_pool_quality.py

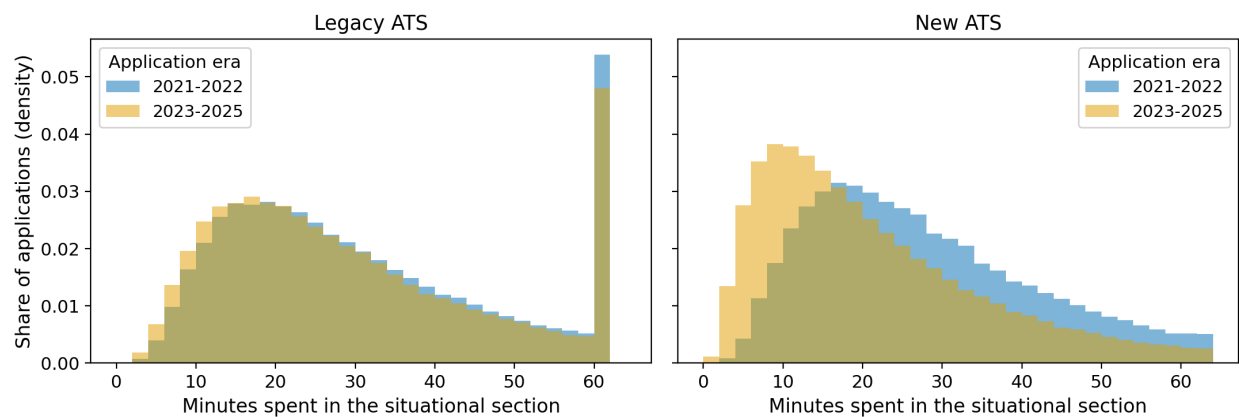
E06c. Essay-score inflation: new ATS x post-AI-era interaction

term	estimate	SE	CI low	CI high	p	n	clusters
new ATS effect, pre-AI era (2021-22)	0.893	0.033	0.828	0.958	<0.001	529,302	40
DiD interaction: extra new-ATS effect, AI era	0.787	0.108	0.575	1	<0.001	529,302	40
post-era gap, new vs legacy (sum of the two)	1.68	0.114	1.457	1.903	<0.001	529,302	40

Sample: n=529302, 40 center clusters. *Notes:* Outcome: essay_rubric_score (0-30). Controls: resume/license/experience/referral/employed. FE: center, application_month. Cluster: center. *Source:* analysis/03_pool_quality.py

E07. Essay minutes distribution

E07. Time spent on the essay collapses on the new ATS after 2023; legacy shifts only slightly



Sample: 529,302 reviewed applications, split by ATS and by application era.
Source: analysis/03_pool_quality.py

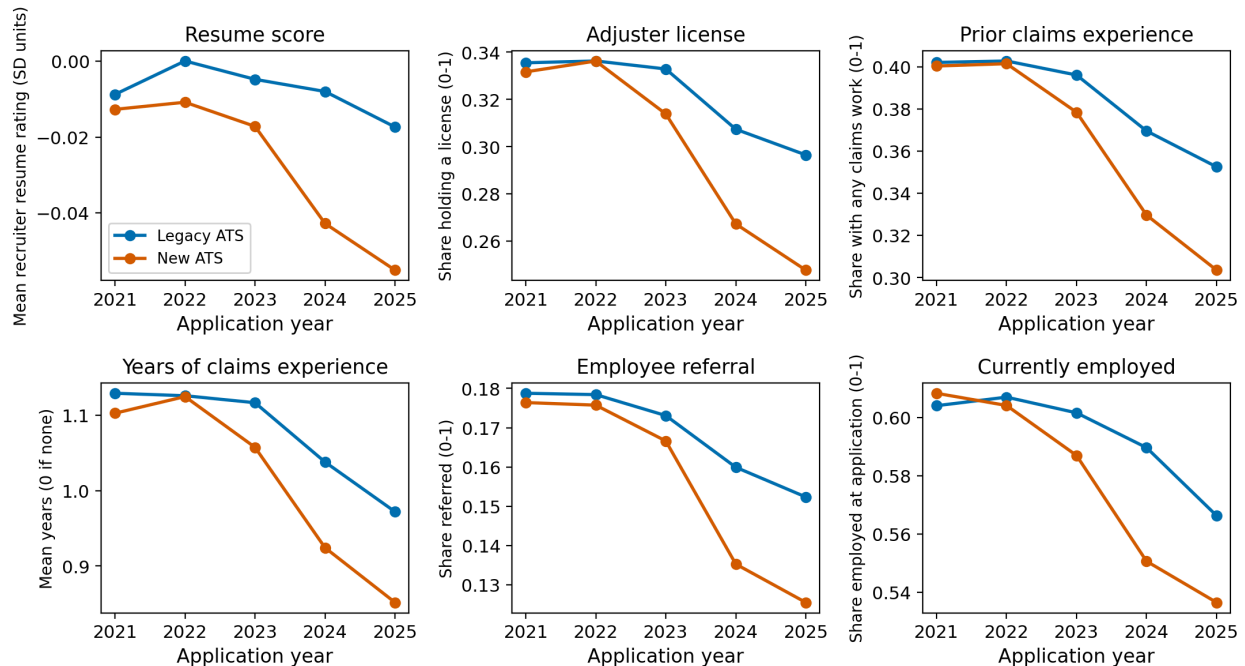
E07b. Mean essay score and share of essays under 10 minutes, by year and ATS

year	('legacy', 'essay')	('legacy', 'fast10')	('new', 'essay')	('new', 'fast10')
2021	17.15	0.061	18.02	0.059
2022	17.16	0.062	18.01	0.063
2023	17.33	0.070	18.46	0.108
2024	17.75	0.094	19.87	0.222
2025	18.05	0.117	20.69	0.283

Sample: 529,302 reviewed applications. *Notes:* The legacy (timed, no-paste) system also drifts after 2023, more slowly: a timed field slows the leak, it does not stop it. *Source:* analysis/03_pool_quality.py

E08. Objective attributes over time

E08. Objective applicant attributes drift slightly DOWN on the new ATS after 2023 -- the pool did not objectively improve while essay scores rose



Sample: 529,302 reviewed applications, yearly means by ATS at application.
Source: analysis/03_pool_quality.py

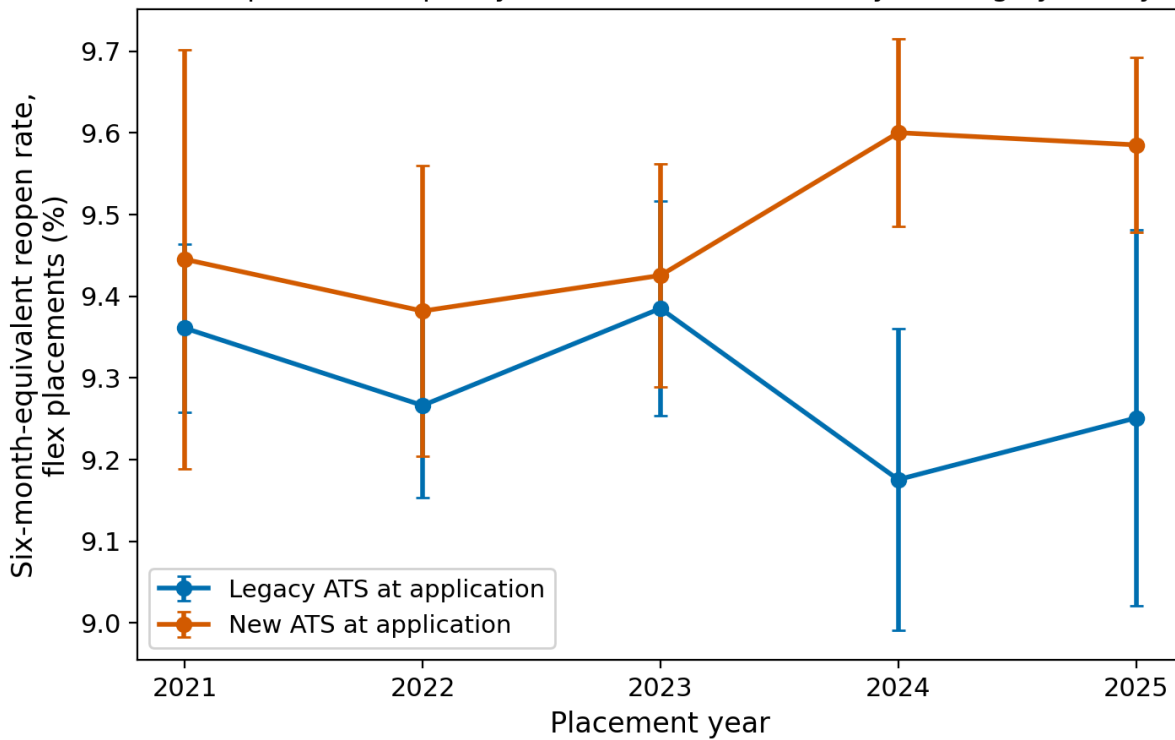
E08b. Objective applicant attributes: new ATS x post-AI-era DiD

outcome	term	estimate	SE	CI low	CI high	p	n	clusters
resume_score	new ATS effect, pre-AI era (2021-22)	-0.003	0.006	-0.015	0.008	0.559	529,302	40
resume_score	DiD interaction: extra new-ATS effect, AI era	-0.010	0.006	-0.021	0.001	0.078	529,302	40
resume_score	post-era gap, new vs legacy (sum of the two)	-0.013	0.007	-0.027	0.000	0.057	529,302	40
has_adjuster_license	new ATS effect, pre-AI era (2021-22)	-0.000	0.003	-0.006	0.006	0.916	529,302	40
has_adjuster_license	DiD interaction: extra new-ATS effect, AI era	-0.026	0.004	-0.035	-0.018	<0.001	529,302	40
has_adjuster_license	post-era gap, new vs legacy (sum of the two)	-0.026	0.004	-0.034	-0.019	<0.001	529,302	40
prior_claims_experience	new ATS effect, pre-AI era (2021-22)	0.001	0.002	-0.004	0.005	0.715	529,302	40
prior_claims_experience	DiD interaction: extra new-ATS effect, AI era	-0.024	0.004	-0.032	-0.015	<0.001	529,302	40
prior_claims_experience	post-era gap, new vs legacy (sum of the two)	-0.023	0.004	-0.030	-0.016	<0.001	529,302	40
prior_claims_years	new ATS effect, pre-AI era (2021-22)	-0.006	0.009	-0.024	0.011	0.469	529,302	40
prior_claims_years	DiD interaction: extra new-ATS effect, AI era	-0.053	0.013	-0.078	-0.028	<0.001	529,302	40
prior_claims_years	post-era gap, new vs legacy (sum of the two)	-0.060	0.011	-0.080	-0.039	<0.001	529,302	40
referral	new ATS effect, pre-AI era (2021-22)	-0.004	0.002	-0.007	-0.000	0.043	529,302	40
referral	DiD interaction: extra new-ATS effect, AI era	-0.011	0.003	-0.017	-0.006	<0.001	529,302	40
referral	post-era gap, new vs legacy (sum of the two)	-0.015	0.003	-0.021	-0.009	<0.001	529,302	40
currently_employed	new ATS effect, pre-AI era (2021-22)	0.003	0.003	-0.003	0.009	0.342	529,302	40
currently_employed	DiD interaction: extra new-ATS effect, AI era	-0.021	0.003	-0.027	-0.015	<0.001	529,302	40
currently_employed	post-era gap, new vs legacy (sum of the two)	-0.018	0.004	-0.025	-0.011	<0.001	529,302	40

Sample: n=529,302 reviewed applications, 40 center clusters. Notes: Each outcome regressed on treated_app + treated_app:post_ai with center and application-month FE, cluster center. Compare with the essay-score DiD in E06c. Source: analysis/03_pool_quality.py

E09. Flex reopen over time

E09. Flex-placement quality (unselected on the essay) is roughly flat by ATS



Sample: 28,672 flex placements, 2021-01 to 2025-12. Error bars: 95% CI.
Source: analysis/03_pool_quality.py

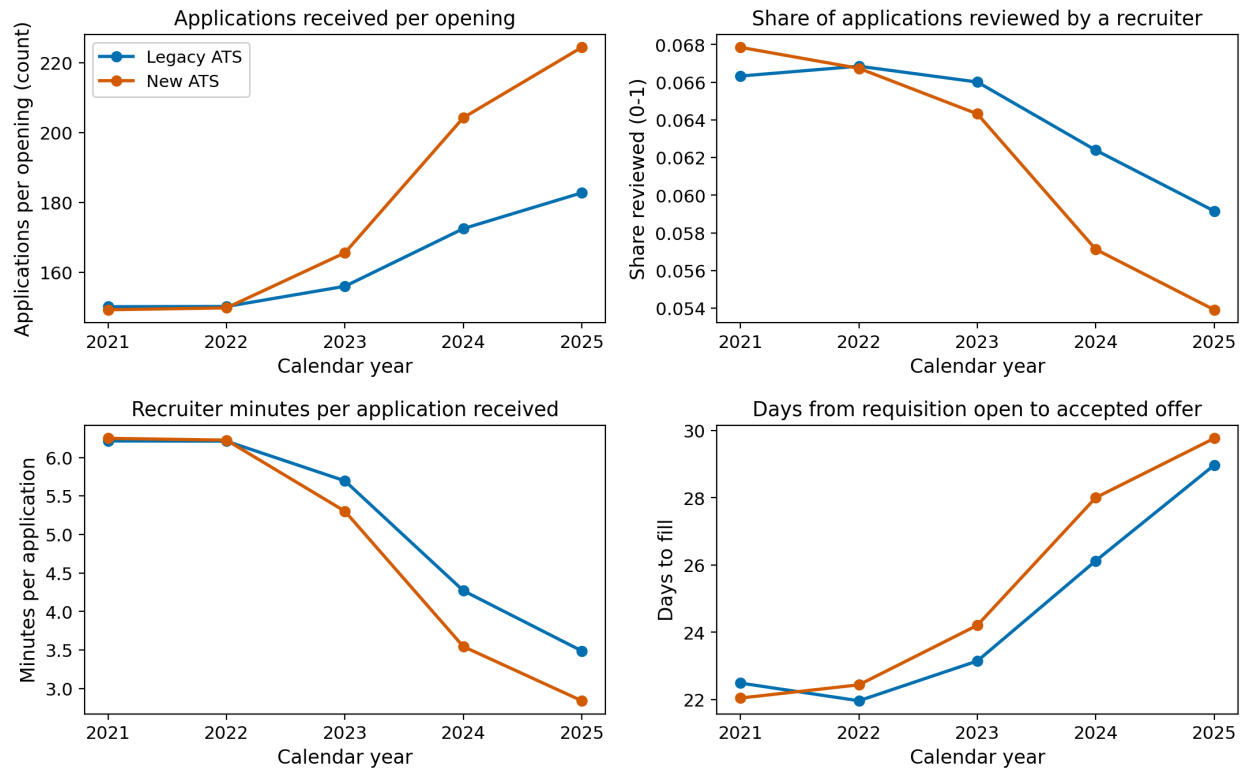
E09b. Flex reopen rate: new ATS x post-AI-era interaction

term	estimate	SE	CI low	CI high	p	n	clusters
new ATS effect, pre-AI era (2021-22)	0.006	0.125	-0.239	0.252	0.961	28,672	40
DiD interaction: extra new-ATS effect, AI era	-0.075	0.136	-0.342	0.192	0.581	28,672	40
post-era gap, new vs legacy (sum of the two)	-0.069	0.085	-0.236	0.098	0.418	28,672	40

Sample: n=28672, 40 center clusters. Notes: Outcome: reopen_rate_pct. Controls: certification_level FE. FE: center, placement_month. Cluster: center. Source: analysis/03_pool_quality.py

E10. Recruiter capacity panel

E10. Recruiter capacity has not kept pace with application volume:
more applications per opening, less time per application, longer to fill



Sample: 2,400 center-months, 40 centers, 2021-01 to 2025-12, yearly means.
Source: analysis/04_recruiter_capacity.py

E10b. Recruiter-capacity metrics by year and ATS

year	ats	apps per opening	review share	recruiter minutes per application	days to fill
2021	legacy	150.2	0.066	6.21	22.49
2021	new	149.3	0.068	6.25	22.04
2022	legacy	150.3	0.067	6.21	21.96
2022	new	149.9	0.067	6.22	22.44
2023	legacy	156	0.066	5.7	23.15
2023	new	165.6	0.064	5.3	24.21
2024	legacy	172.5	0.062	4.27	26.13
2024	new	204.3	0.057	3.54	28
2025	legacy	182.8	0.059	3.49	28.97
2025	new	224.4	0.054	2.84	29.77

Sample: 2,400 center-months, yearly averages by ATS in that center-month. Source: analysis/04_recruiter_capacity.py

E10c. Drivers of days-to-fill

	estimate	SE	CI low	CI high	p	n	clusters
Intercept	21.37	3.894	13.74	29.01	<0.001	2,400	40
apps_per_opening	0.028	0.011	0.006	0.049	0.011	2,400	40
recruiter_minutes_per_application	-0.792	0.384	-1.544	-0.040	0.039	2,400	40
local_unemployment_rate	0.537	0.275	-0.002	1.075	0.051	2,400	40

Sample: n=2400, 40 center clusters. *Notes:* Outcome: days_to_fill. FE: center, month. Cluster: center. *Source:* analysis/04_recruiter_capacity.py

E10d. Applications received per center-month: new ATS x AI-era DiD

term	estimate	SE	CI low	CI high	p	n	clusters
new ATS effect, pre-AI era (2021-22)	-478.3	91	-656.7	-300	<0.001	2,400	40
DiD interaction: extra new-ATS effect, AI era	826.1	117.5	595.8	1,056	<0.001	2,400	40
post-era gap, new vs legacy (sum of the two)	347.8	126.8	99.3	596.3	0.010	2,400	40

Sample: n=2400 center-months, 40 clusters. *Notes:* FE: center, month. Cluster: center. *Source:* analysis/04_recruiter_capacity.py

E10e. Recruiter-capacity metrics by year, all 40 centers pooled

year	apps per opening	review share	recruiter minutes per application	days to fill
2021	150.1	0.067	6.22	22.43
2022	150.1	0.067	6.22	22.1
2023	160.6	0.065	5.51	23.65
2024	195.9	0.059	3.73	27.51
2025	217.1	0.055	2.95	29.63

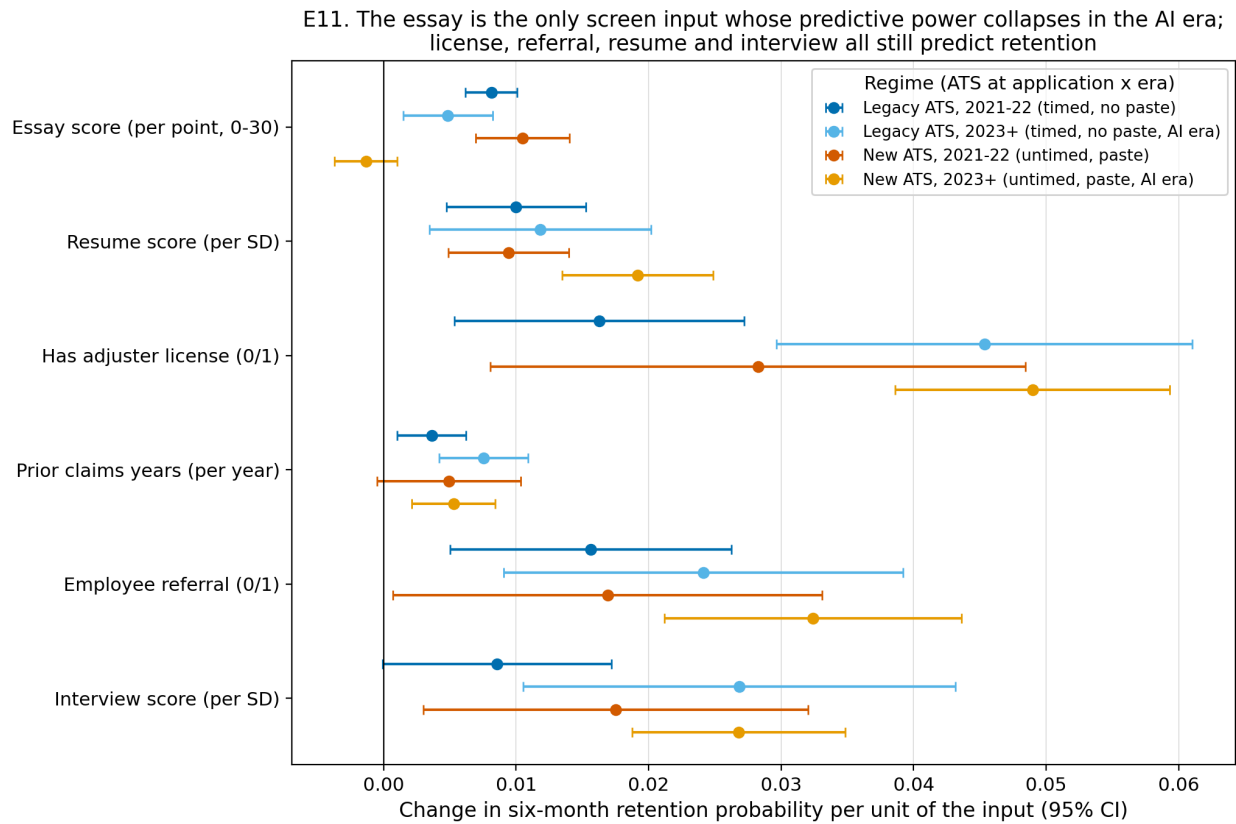
Sample: 2,400 center-months, yearly averages across all centers regardless of ATS. *Notes:* Pooled version of E10b; the narrative quotes these all-center figures. *Source:* analysis/04_recruiter_capacity.py

E10f. Recruiter minutes per application vs applications per opening

	estimate	SE	CI low	CI high	p	n	clusters
Intercept	9.73	0.072	9.589	9.872	<0.001	2,400	40
apps_per_opening	-0.023	0.000	-0.024	-0.022	<0.001	2,400	40

Sample: n=2400 center-months, 40 center clusters. *Notes:* Outcome: recruiter_minutes_per_application. FE: center, month. Cluster: center. A negative coefficient means each extra applicant per opening buys less review time per application: recruiters spread a fixed capacity over a growing queue. *Source:* analysis/04_recruiter_capacity.py

E11. Screen component coefficients by regime



Sample: 41,677 hires with an observed six-month outcome, split into four cells (legacy/new ATS x application before/after 2023-01); 4,589 censored hires excluded.
Source: analysis/05_screen_components.py

E11b. Predictors of six-month retention, by regime

regime	predictor	estimate	SE	CI low	CI high	n	clusters
legacy_pre	essay_rubric_score	0.008	0.001	0.006	0.010	13,040	40
legacy_pre	resume_score	0.010	0.003	0.005	0.015	13,040	40
legacy_pre	has_adjuster_license	0.016	0.006	0.005	0.027	13,040	40
legacy_pre	prior_claims_years	0.004	0.001	0.001	0.006	13,040	40
legacy_pre	referral	0.016	0.005	0.005	0.026	13,040	40
legacy_pre	interview_score	0.009	0.004	-0.000	0.017	13,040	40
legacy_post	essay_rubric_score	0.005	0.002	0.002	0.008	6,326	28
legacy_post	resume_score	0.012	0.004	0.004	0.020	6,326	28
legacy_post	has_adjuster_license	0.045	0.008	0.030	0.061	6,326	28
legacy_post	prior_claims_years	0.008	0.002	0.004	0.011	6,326	28
legacy_post	referral	0.024	0.008	0.009	0.039	6,326	28
legacy_post	interview_score	0.027	0.008	0.011	0.043	6,326	28
new_pre	essay_rubric_score	0.011	0.002	0.007	0.014	5,920	12
new_pre	resume_score	0.009	0.002	0.005	0.014	5,920	12
new_pre	has_adjuster_license	0.028	0.010	0.008	0.048	5,920	12
new_pre	prior_claims_years	0.005	0.003	-0.001	0.010	5,920	12
new_pre	referral	0.017	0.008	0.001	0.033	5,920	12
new_pre	interview_score	0.018	0.007	0.003	0.032	5,920	12
new_post	essay_rubric_score	-0.001	0.001	-0.004	0.001	16,391	33
new_post	resume_score	0.019	0.003	0.013	0.025	16,391	33
new_post	has_adjuster_license	0.049	0.005	0.039	0.059	16,391	33
new_post	prior_claims_years	0.005	0.002	0.002	0.008	16,391	33
new_post	referral	0.032	0.006	0.021	0.044	16,391	33

regime	predictor	estimate	SE	CI low	CI high	n	clusters
new_post	interview_score	0.027	0.004	0.019	0.035	16,391	33

Sample: 41,677 hires with an observed outcome, split into four ATS x era cells. *Notes:* Outcome: retained_6mo (linear probability). FE: center, start_month. Cluster: center. Each cell is a separate regression on its subsample; all six predictors enter together. *Source:* analysis/05_screen_components.py

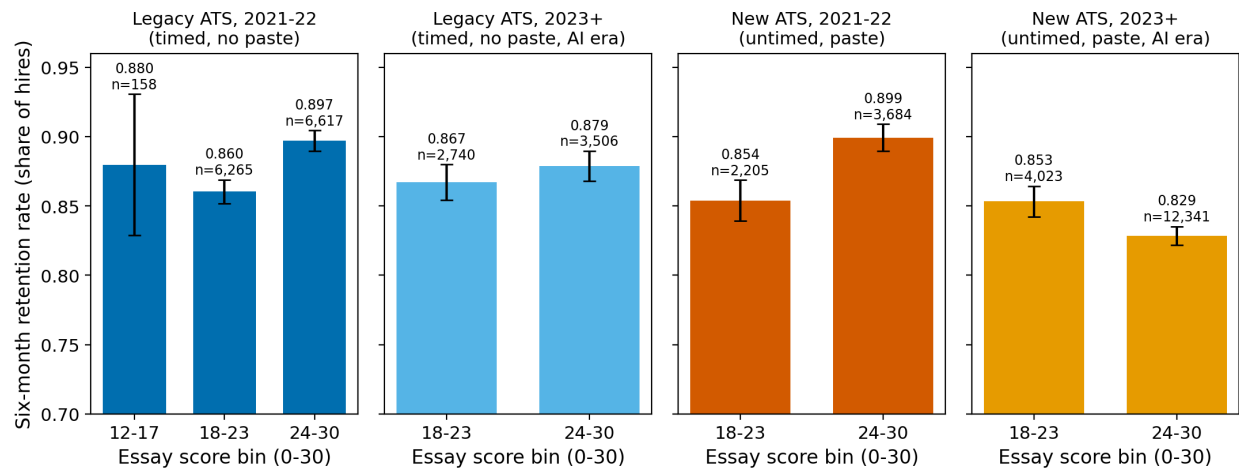
E11c. How much weight recruiters place on the essay score, by regime

regime	essay weight in screen score	SE	CI low	CI high	n	r squared
legacy_pre	0.250	0.000	0.250	0.251	135,320	0.862
legacy_post	0.250	0.001	0.249	0.251	76,215	0.848
new_pre	0.250	0.000	0.249	0.250	62,232	0.861
new_post	0.249	0.000	0.249	0.250	255,535	0.821

Sample: All reviewed applications, split by regime. *Notes:* Outcome: recruiter_screen_score. FE: center, application_month. Cluster: center. An R-squared near 1 means the screen score is a fixed formula of its inputs, not a judgment. *Source:* analysis/05_screen_components.py

E12. Retention by essay bin regime

E12. A top-bin essay (24-30) used to predict better retention than an 18-23 essay; on the new ATS after 2023 it predicts slightly worse retention



Sample: 41,677 permanent hires with an observed six-month outcome, binned by essay score within four ATS x era cells; bins with under 100 hires not shown. Error bars: 95% CI. Y-axis starts at 0.70. *Source:* analysis/06_essay_validity.py

E12b. Essay-score slope on six-month retention, by regime (hires)

regime	estimate	SE	CI low	CI high	n cell
legacy_pre	0.009	0.001	0.007	0.010	13,040
legacy_post	0.005	0.002	0.001	0.008	6,326
new_pre	0.011	0.002	0.007	0.014	5,920
new_post	-0.003	0.001	-0.005	-0.000	16,391

Sample: n=41,677, 40 center clusters. *Notes:* Outcome: retained_6mo. Controls: resume, license, prior years, referral, interview score. FE: center, start_month. Cluster: center. Hires are selected on the essay, which attenuates every slope toward zero. *Source:* analysis/06_essay_validity.py

E12c. Essay-score slope on six-month reopen rate, by regime (hires)

regime	estimate	SE	CI low	CI high	n cell
legacy_pre	-0.147	0.010	-0.167	-0.127	13,040
legacy_post	-0.093	0.015	-0.123	-0.063	6,326
new_pre	-0.180	0.016	-0.212	-0.148	5,920
new_post	0.035	0.010	0.016	0.054	16,391

Sample: n=41,677, 40 center clusters. *Notes:* Outcome: reopen_rate_6mo_pct (percentage points).

Negative = higher essay score predicts fewer reopens. Same FE/cluster as E12b. *Source:* analysis/06_essay_validity.py

E12d. Essay-score slope on reopen rate, by regime (flex placements -- NOT selected on the essay by Meridian)

regime	estimate	SE	CI low	CI high	n cell	weighted estimate	weighted SE
legacy_pre	-0.179	0.010	-0.199	-0.160	9,079	-0.177	0.010
legacy_post	-0.122	0.012	-0.146	-0.098	5,541	-0.118	0.013
new_pre	-0.207	0.019	-0.244	-0.169	2,346	-0.212	0.021
new_post	-0.069	0.007	-0.083	-0.056	11,706	-0.069	0.007

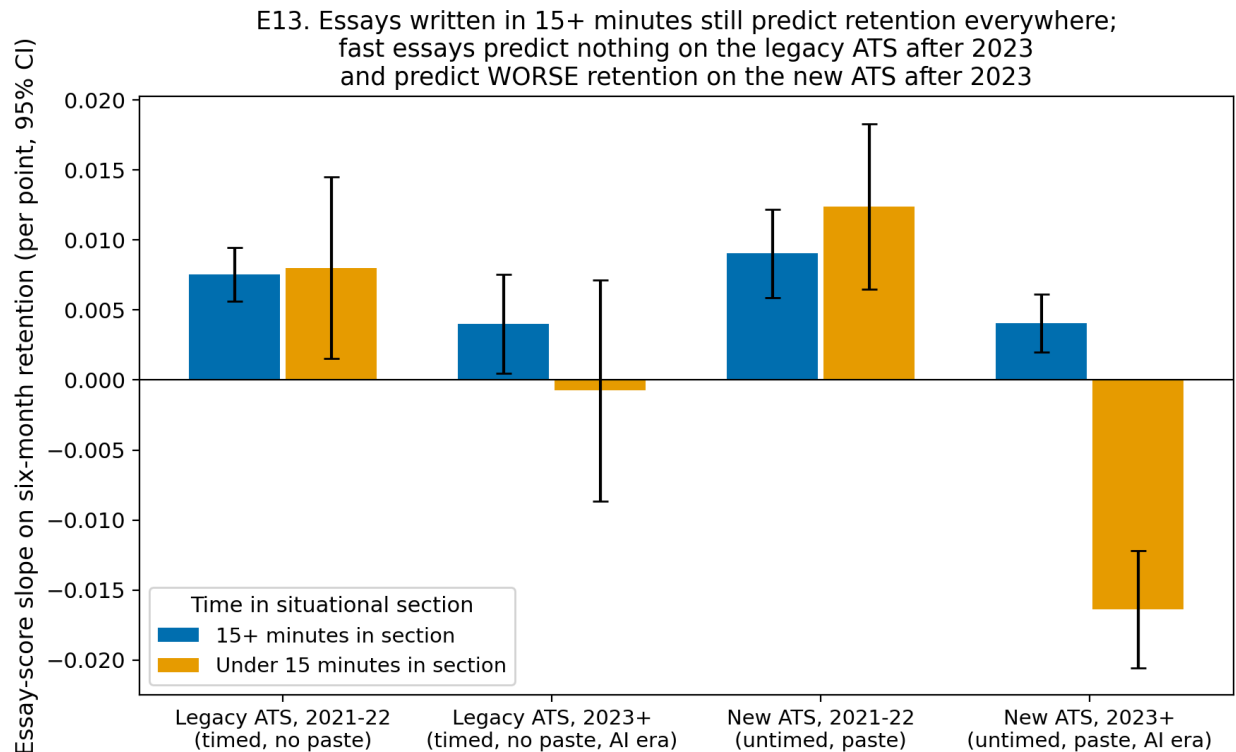
Sample: n=28,672 flex placements, 40 center clusters. *Notes:* Outcome: reopen_rate_pct. Controls: resume, license, prior years, referral, certification level. FE: center, placement_month. Cluster: center. A MORE NEGATIVE slope means the essay is more informative. Weighted columns weight by claims_closed. *Source:* analysis/06_essay_validity.py

E12e. Spec C: essay slope on retention in the 12 months before vs after a center's own ATS migration (2023/24 waves) vs never-migrated centers over the same calendar window

group	window	estimate	SE	CI low	CI high	n
2023/24-wave centers, own migration month	12 months before	0.006	0.002	0.002	0.010	8,327
2023/24-wave centers, own migration month	12 months after	-0.002	0.002	-0.006	0.003	8,327
never-migrated centers, pseudo-event 2023-09	12 months before	0.007	0.003	0.002	0.012	2,359
never-migrated centers, pseudo-event 2023-09	12 months after	-0.003	0.002	-0.007	0.001	2,359

Sample: 8,327 hires at 2023/2024-wave centers; 2,359 hires at never-migrated centers. *Notes:* Same controls/FE/cluster as E12b. The never-migrated slope also falls over this window (calendar 2022-09 to 2024-08), so the migration itself is not the whole story: the AI era degrades the timed section too, more slowly. *Source:* analysis/06_essay_validity.py

E13. Essay slope by speed regime



Sample: 41,677 hires with an observed six-month outcome, split by ATS x era cell and by time spent in the situational section.

Source: analysis/06_essay_validity.py

E13b. Essay-score slope on retention, by regime and by whether the applicant spent under 15 minutes in the situational section

regime	fast lt 15min	estimate	SE	CI low	CI high	n cell
legacy_pre	0	0.008	0.001	0.006	0.009	11,699
legacy_pre	1	0.008	0.003	0.002	0.014	1,341
legacy_post	0	0.004	0.002	0.000	0.008	5,270
legacy_post	1	-0.001	0.004	-0.009	0.007	1,056
new_pre	0	0.009	0.002	0.006	0.012	5,353
new_pre	1	0.012	0.003	0.006	0.018	567
new_post	0	0.004	0.001	0.002	0.006	9,876
new_post	1	-0.016	0.002	-0.021	-0.012	6,515

Sample: 41,677 hires with an observed outcome. Notes: Outcome: retained_6mo. essay x cell x fast15 slopes with cell x fast15 main effects, center + start-month FE, cluster center. fast_lt_15min=1 means essay_section_minutes < 15. Source: analysis/06_essay_validity.py

E13c. Essay score per minute spent in the section, by regime

regime	estimate	SE	n
legacy_pre	0.054	0.001	135,320
legacy_post	0.030	0.003	76,215
new_pre	0.034	0.001	62,232
new_post	-0.021	0.001	255,535

Sample: All reviewed applications, by cell. *Notes:* essay_rubric_score ~ essay_section_minutes + center FE + application-month FE, cluster center. Positive = longer effort earns a higher score. *Source:* analysis/06_essay_validity.py

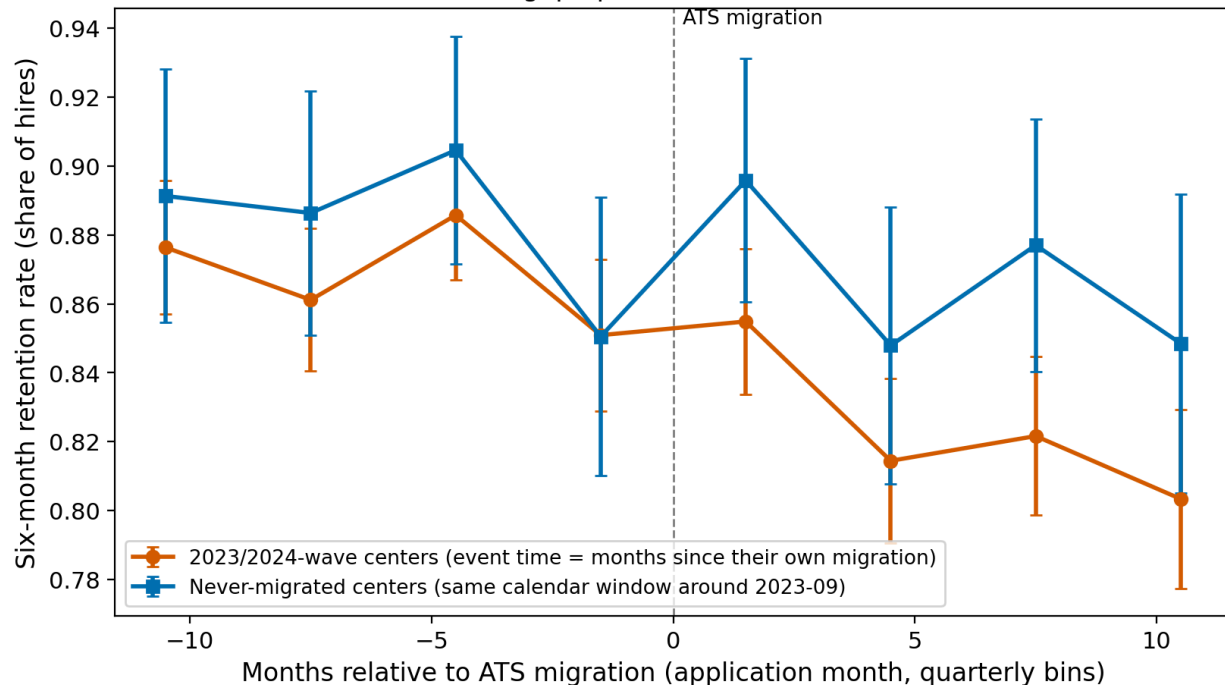
E14. Placebo test: slope of other screen inputs on six-month retention, by regime

regime	estimate	SE	CI low	CI high	variable
legacy_pre	0.009	0.003	0.004	0.014	resume_score
legacy_post	0.012	0.004	0.003	0.020	resume_score
new_pre	0.008	0.002	0.003	0.013	resume_score
new_post	0.023	0.003	0.017	0.028	resume_score
legacy_pre	0.020	0.006	0.009	0.030	has_adjuster_license
legacy_post	0.050	0.009	0.033	0.066	has_adjuster_license
new_pre	0.033	0.010	0.014	0.052	has_adjuster_license
new_post	0.060	0.005	0.050	0.071	has_adjuster_license
legacy_pre	0.004	0.001	0.001	0.007	prior_claims_years
legacy_post	0.009	0.002	0.005	0.012	prior_claims_years
new_pre	0.005	0.003	0.000	0.011	prior_claims_years
new_post	0.008	0.002	0.005	0.011	prior_claims_years
legacy_pre	-0.005	0.004	-0.013	0.004	interview_score
legacy_post	0.019	0.008	0.003	0.035	interview_score
new_pre	0.001	0.007	-0.013	0.014	interview_score
new_post	0.032	0.004	0.024	0.040	interview_score

Sample: 41,677 hires with an observed outcome (interview_score restricted to applicants who were interviewed). *Notes:* Univariate slopes (each input alone) with center + start-month FE, cluster center. If these are stable while the essay's slope collapses (E12b), the essay result is not an artifact of a center-wide shift in hiring quality or measurement. E11b gives the multivariate version. *Source:* analysis/06_essay_validity.py

E15. Migration event study

E15. Migrating and never-migrated centers track each other before migration; the gap opens after the switch



Sample: 8,327 hires at 2023/2024-migration-wave centers (-12 to +11 months around each center's migration) and 2,359 hires at never-migrated centers (same window around the waves' median migration month). Error bars: 95% CI.

Source: analysis/07_identification.py

E15b. Event-study means: retention before/after migration vs never-migrated

period	migrating centers	never migrated	gap never minus migrating
12 months before	0.869	0.883	0.014
12 months after	0.825	0.868	0.043

Sample: 8,327 + 2,359 hires, event window +/-12 months. Source: analysis/07_identification.py

E15c. Selection on the essay into the hire sample, by regime

regime4	applicant essay mean	applicant essay sd	hire essay mean	hire essay sd	hire rate bin 18 23	hire rate bin 24 30	share applicants 24 30
legacy_pre	17.16	4.607	23.65	2.957	0.120	0.579	0.084
legacy_post	17.56	4.74	24	2.975	0.097	0.486	0.105
new_pre	18.02	4.591	24.5	2.869	0.083	0.501	0.118
new_post	19.88	4.832	25.55	2.845	0.041	0.274	0.232

Sample: All reviewed applications and all hires with an observed outcome. Notes: If the hire-sample essay SD were much smaller in one cell, a zero slope there could be a range-restriction artifact. It is not: the SD is similar in every cell. What changed is the share of applicants scoring 24-30 and the hire rate within that bin. Source: analysis/07_identification.py

E15d. Small-cluster inference: wild-cluster bootstrap on the headline essay-slope differences

test	difference	cluster SE	cluster p	wild bootstrap p 299	n	clusters
hires: retention slope, new_post minus legacy_pre	-0.011	0.001	0.000	<0.001	41,677	40
hires: retention slope, legacy_post minus legacy_pre	-0.004	0.002	0.063	0.080	41,677	40
flex: reopen slope, new_post minus legacy_pre	0.110	0.013	0.000	<0.001	28,672	40
flex: reopen slope, legacy_post minus legacy_pre	0.057	0.016	0.000	0.003	28,672	40

Sample: Hires with an observed outcome (retention) and flex placements (reopen). *Notes:* H0: the essay slope in the named cell equals the legacy pre-2023 slope. Rademacher wild cluster bootstrap, 299 draws, 40 center clusters, restricted model imposes H0. A p below 0.05 means the difference survives small-cluster inference. *Source:* analysis/07_identification.py

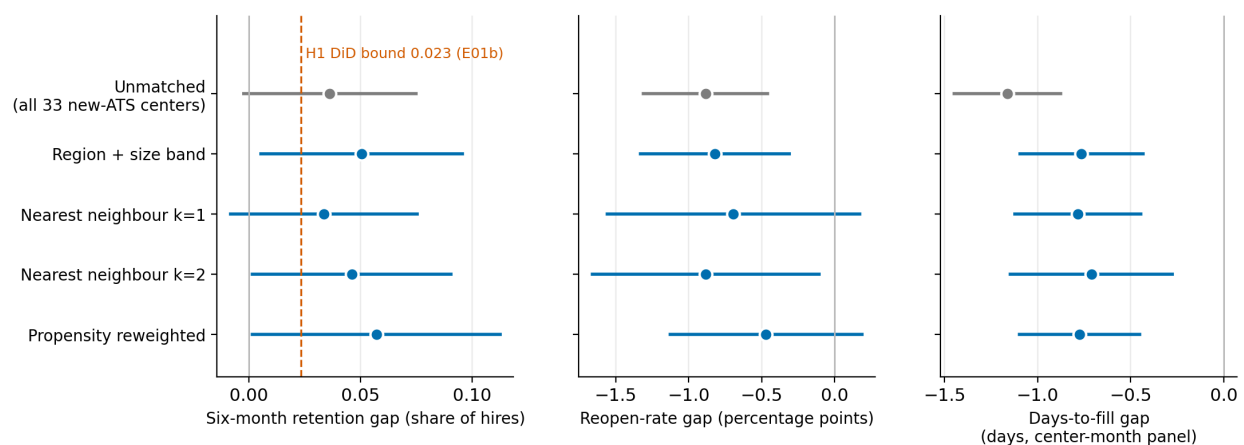
E15e. Share of the legacy pre-2023 essay signal retained, by cell: hires sample vs flex sample

regime	hires share of legacy pre slope retained	flex share of legacy pre slope retained	range retained
legacy_pre	1	1	[100%, 100%]
legacy_post	0.547	0.681	[55%, 68%]
new_pre	1.232	1.153	[115%, 123%]
new_post	0.000	0.387	[0%, 39%]

Sample: Slopes from E12b (hires) and E12d (flex). Negative hire slopes are clipped at 0% retained. *Notes:* The two samples bracket the truth: hires are selected on the essay (attenuated, lower bound on retained signal); flex are not. *Source:* analysis/07_identification.py

E15f. Matched legacy comparison

E15f. The legacy centers' retention advantage does not shrink once matched on size and region (unmatched 0.036, matched 0.046 [0.001, 0.091] with k=2 nearest neighbours), but 7 vs 5-19 centers is thin. All gaps are legacy minus new-ATS, 2024-25.



Sample: 1,737 hires at 7 never-migrated centers vs 11,440 new-ATS hires at 33 centers, starting 2024-01 or later, observed six-month outcome; days to fill from 168 + 750 center-months. Bars: 95% CI, SEs clustered by center (12-40 clusters; unreliable). *Source:* analysis/07_identification.py

E15f. Matched comparison: legacy-minus-new-ATS gaps in 2024-25, unmatched vs matched on size and region

approach	outcome	unmatched gap	matched gap	matched CI low	matched CI high	wild boot p	matched centers	matched rows
band	retained_6mo	0.036	0.051	0.005	0.096	0.074	9	2050
band	reopen_rate_6mo_pct	-0.884	-0.821	-1.331	-0.311	—	9	2050
band	days_to_fill	-1.162	-0.765	-1.095	-0.434	—	9	198
nn1	retained_6mo	0.036	0.034	-0.008	0.075	—	5	1,342
nn1	reopen_rate_6mo_pct	-0.884	-0.694	-1.56	0.172	—	5	1,342
nn1	days_to_fill	-1.162	-0.784	-1.123	-0.446	—	5	118
nn2	retained_6mo	0.036	0.046	0.001	0.091	—	7	1984
nn2	reopen_rate_6mo_pct	-0.884	-0.884	-1.661	-0.107	—	7	1984
nn2	days_to_fill	-1.162	-0.712	-1.148	-0.276	—	7	166
ps	retained_6mo	0.036	0.057	0.002	0.113	—	19	6,792
ps	reopen_rate_6mo_pct	-0.884	-0.470	-1.128	0.188	—	19	6,792
ps	days_to_fill	-1.162	-0.775	-1.098	-0.453	—	19	436

Sample: Legacy = 7 never-migrated centers, 1,737 hires starting 2024-01 or later with an observed six-month outcome (168 center-months for days to fill). Comparison = new-ATS-at-application hires starting 2024-01 or later (unmatched: 33 centers, 11,440 hires; matched: see n columns). *Notes:* Gap = coefficient on a legacy dummy in outcome ~ legacy + start-month (or month) FE, linear probability model for retention, SEs clustered by center; center FE are not identified for a between-center contrast. Approaches: band = comparison centers in the legacy regions (Mid-Atlantic, Midwest, Southeast) with size_index in [0.511, 1.088] (the legacy centers' min..max, no margin), unweighted; nn1/nn2 = each legacy center matched with replacement to its 1 or 2 nearest new-ATS centers on size_index within the same region, comparison centers weighted by times matched, spread evenly over their hires; ps = logit P(legacy | size_index, region) on the centers in legacy regions, comparison weight p/(1-p). Matched comparison centers, as center(weight): band: C00(1), C01(1), C05(1), C07(1), C14(1), C19(1), C25(1), C27(1), C32(1); nn1: C05(1), C12(1), C19(1), C25(3), C27(1); nn2: C01(1), C05(1), C07(2), C12(1), C19(4), C25(4), C27(1); ps: C00(1.1), C01(0.081), C02(0.0015), C03(0.12), C05(0.42), C07(0.32), C11(0.013), C12(0.24), C13(0.1), C14(0.42), C19(0.91), C22(0.13), C24(0.0025), C25(2.4), C27(0.22), C28(0.014), C32(0.42), C35(0.056), C36(0.039). Unmatched 95% CIs, legacy n and per-approach detail are in E15f_matched_comparison_full.csv. With 12-26 clusters the cluster-robust CIs are unreliable; the wild-cluster bootstrap p (299 Rademacher draws) is reported for the unweighted band retention gap only. The H1 DiD bound (E01b) is 0.0232. The '(b, matched)' row in E19 uses the nn2 retention and reopen gaps. *Source:* analysis/07_identification.py

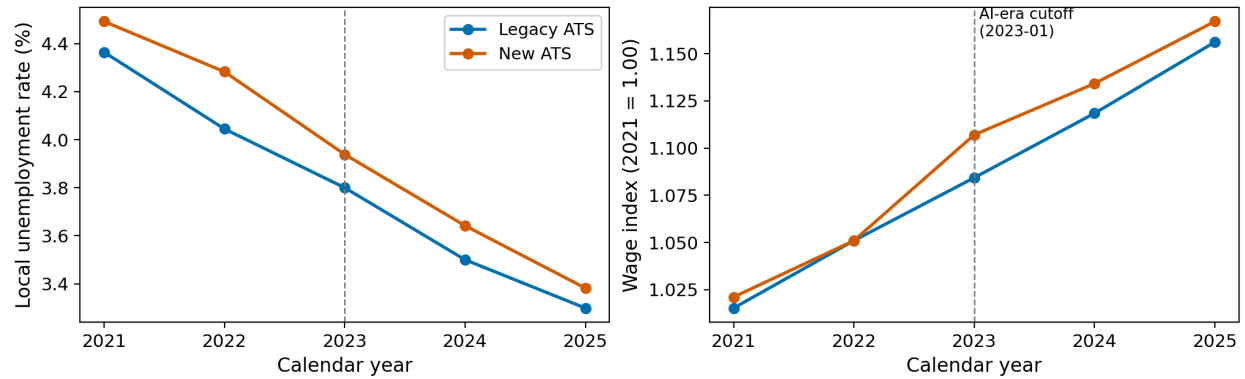
E15g. Pre-2023 essay slope on retention: pilot centers vs never-migrated centers

centers	essay slope on retention	SE cluster center	CI low	CI high	n hires	clusters
2021 vendor pilot (migrated 2021)	0.010	0.001	0.007	0.012	8,012	12
Never migrated (legacy throughout)	0.008	0.002	0.005	0.012	2,393	7

Sample: Hires applying before 2023-01 with an observed six-month outcome: 8,012 at pilot centers, 2,393 at never-migrated centers. *Notes:* retained_6mo ~ essay_rubric_score + controls, center FE + start-month FE, SEs clustered by center. If migration had selected centers where the essay was more or less valid, these slopes would differ. *Source:* analysis/07_identification.py

E16. Labor market by ATS

E16. Local labor-market conditions move together regardless of ATS;
they do not explain the new-vs-legacy divergence



Sample: 2,400 center-months, 40 centers, yearly means by ATS in that center-month; dashed line = 2023 AI-era cutoff.

Source: analysis/07_identification.py

E16b. Labor-market conditions by year and ATS

year	ats	local unemployment rate	wage index
2021	legacy	4.364	1.015
2021	new	4.492	1.021
2022	legacy	4.045	1.051
2022	new	4.283	1.051
2023	legacy	3.8	1.084
2023	new	3.939	1.107
2024	legacy	3.5	1.118
2024	new	3.642	1.134
2025	legacy	3.299	1.156
2025	new	3.382	1.167

Sample: 2,400 center-months. Source: analysis/07_identification.py

E17. Cost per retained employee: status quo vs counterfactuals where the essay still screened

scenario	n	r	days to fill	reopen %	C hire	C sep	C policy	C quality	CPR excl. quality	CPR incl. quality
Status quo: new ATS, hires starting 2024-25 (observed)	11,440	0.820	28.93	8.918	18,699	2,540	0.000	17,531	23,377	40,908
Status quo (alt): new ATS, all AI-era hires 2023-25 (observed)	16,391	0.834	28.93	8.592	18,699	2,540	0.000	16,888	22,916	39,804
Counterfactual A: legacy centers, hires starting 2024-25 (observed)	2,487	0.857	27.26	7.985	18,334	2,540	0.000	15,695	21,821	37,516
Counterfactual A': status quo + DiD retention gain, status-quo days to fill	—	0.843	28.93	7.985	18,699	2,540	0.000	15,695	22,665	38,360
Counterfactual B: new ATS before the AI era (2021-22), costed at 2025 rates	5,920	0.882	28.93	7.645	18,699	2,540	0.000	15,027	21,547	36,574

Sample: Hires with an observed six-month outcome, by scenario; all costed at 2025 average rates (E17a). CPR = cost per retained employee, \$. *Notes:* A = the seven never-migrated centers (different, smaller centers). A' = the status-quo cohort with retention raised by the H1 DiD gain (E01b) and reopen set to the legacy level. B = the same new-ATS centers before AI writing. C_hire excludes recruiter screening time (E17b). *Source:* analysis/08_cost_worksheet.py

E17a. Worksheet inputs: 2025 average rates across 40 centers vs the prompt's worked example

	2025 average	prompt worked example
sourcing_cost_per_opening	746.9	747
training_cost_per_hire	5,085	5,085
ramp_productivity_loss_per_hire	6,540	6,540
vacancy_cost_per_day	218.7	219
separation_cost_per_exit	2,540	2,540
claims_per_adjuster_6mo	608.5	609
cost_per_reopened_claim	323	323
recruiter_hourly_rate	66.53	—

Sample: 480 center-months in 2025. Source: analysis/08_cost_worksheet.py

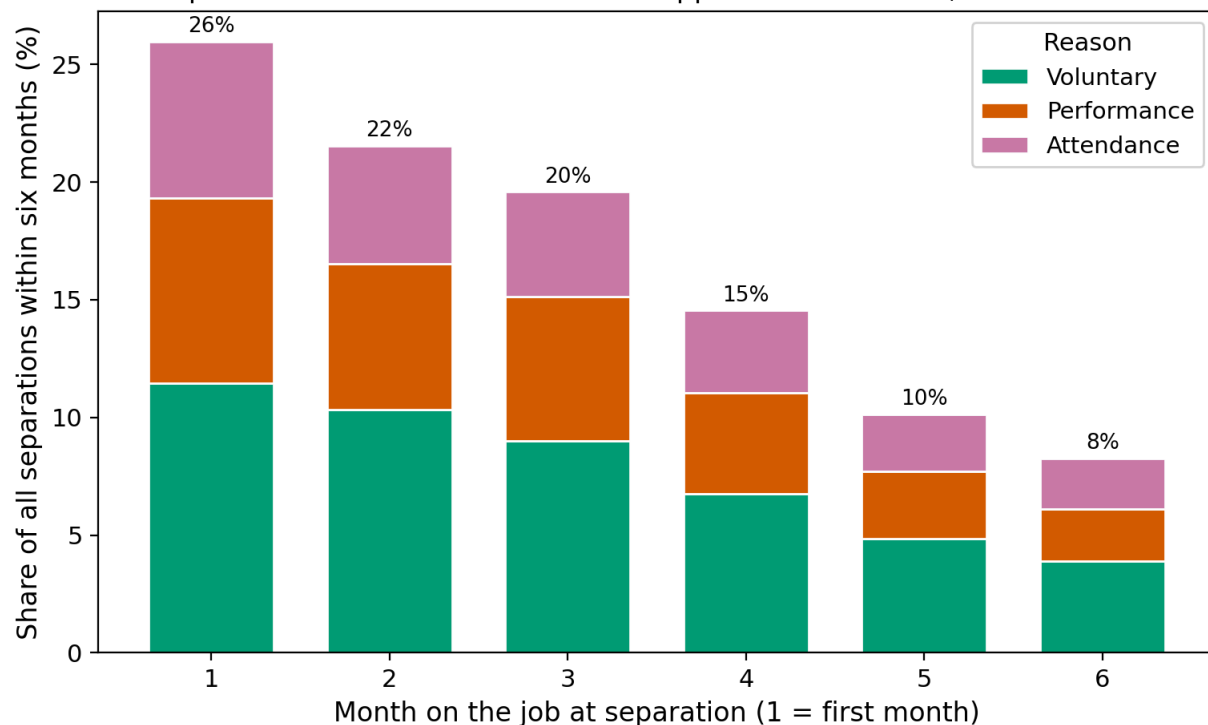
E17b. Recruiter screening time per hire (not in the prompt's C_hire; shown for option c)

	value
recruiter_minutes_per_application_2025	2.954
applications_received_per_hire_2025 (all centers, annualised)	226
recruiter_hourly_rate_2025	66.53
recruiter_screening_cost_per_hire_\$	740.4
as_share_of_C_hire_status_quo	0.040

Sample: 2025 center-months and 2025 hires (H1 x 2). Source: analysis/08_cost_worksheet.py

E18. Separation hazard by month

E18. Separations are front-loaded: 47% happen in months 1-2, 33% in months 4-6



Sample: 5,796 hires who separated within six months, out of 41,677 with an observed outcome (all regimes).

Source: analysis/08_cost_worksheet.py

E18b. Separations by month on the job and reason (counts)

months to sep	attendance	performance	voluntary	total
1	385	457	663	1,505
2	291	359	598	1,248
3	258	355	522	1,135
4	203	249	391	843
5	140	166	281	587
6	124	129	225	478

Sample: 5,796 hires who separated within six months. *Source:* analysis/08_cost_worksheet.py

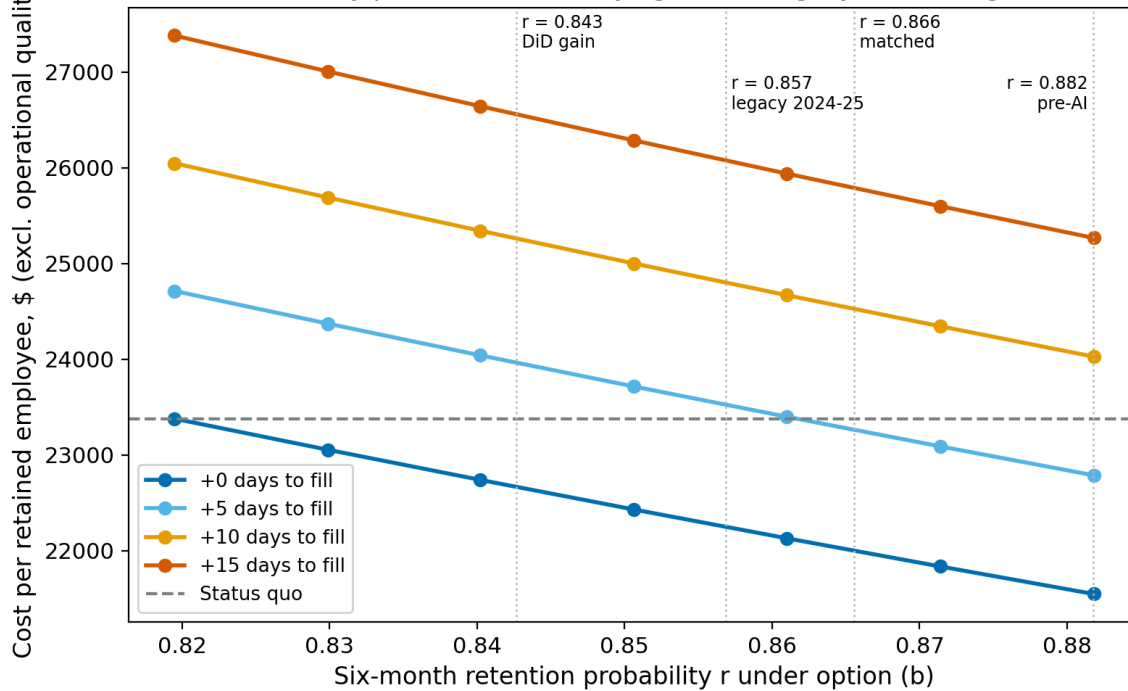
E19. Cost per retained employee by option, vs status quo

scenario	n	r	days to fill	reopen %	C hire	C sep	C policy	C quality	CPR excl. quality	CPR incl. quality	delta vs status quo excl. quality	annual delta excl. quality \$M	delta vs status quo incl. quality	annual delta incl. quality \$M
Status quo: new ATS, hires starting 2024-25 (observed)	11,440	0.820	28.93	8.918	18,699	2,540	0.000	17,531	23,377	40,908	0.000	0.000	0.000	0.000
(a) Do nothing: status quo continues	11,440	0.820	28.93	8.918	18,699	2,540	0.000	17,531	23,377	40,908	0.000	0.000	0.000	0.000
(a') Do nothing, 2024->25 retention change (+0.000) continues one more year	—	0.820	28.93	8.918	18,699	2,540	0.000	17,531	23,354	40,885	-22.88	-0.160	-22.88	-0.160
(b, low) Applicants bear it: retention gain = H1 DiD only; reopen = legacy	—	0.843	28.93	7.985	18,699	2,540	0.000	15,695	22,665	38,360	-712.6	-4.87	-2,548	-17.4
(b) Applicants bear it: timed/no-paste section; r and reopen = legacy centers 2024-25; status-quo days to fill	2,487	0.857	28.93	7.985	18,699	2,540	0.000	15,695	22,247	37,942	-1,130	-7.72	-2,966	-20.26
(b, high) Applicants bear it: legacy r, reopen AND legacy days to fill	2,487	0.857	27.26	7.985	18,334	2,540	0.000	15,695	21,821	37,516	-1,556	-10.63	-3,392	-23.17
(b, matched) Applicants bear it: r and reopen = status quo + legacy gap matched on size and region (nearest neighbour, k=2, on size_index within region, E15f); status-quo days to fill	—	0.866	28.93	8.035	18,699	2,540	0.000	15,794	21,998	37,792	-1,379	-9.42	-3,116	-21.28
(c) Meridian bears it: structured interview; ASSUMED r recovers 50% of the legacy gap, +15 recruiter min per reviewed application (\$204/hire), +3 days to fill	—	0.838	31.93	8.452	19,559	2,540	0.000	16,613	23,826	40,439	448.5	3.06	-469.4	-3.21
(d) Bear it after hiring: \$500 retention bonus at six months; ASSUMED r recovers 17% of the legacy gap (separations are front-loaded, E18)	—	0.826	28.93	8.918	18,699	2,540	500	17,531	23,682	41,213	304.5	2.08	304.5	2.08

Sample: Status quo, (a) and (b) are observed cohorts; (a'), (b low/high/matched), (c) and (d) rest on the stated assumptions in the row label. Annual figures use 6,830 retained seats a year (8,334 new-ATS hires x r, 2025 H1 pace x 2). *Notes:* Negative delta = cheaper than status quo. (c) and (d) are not estimated from Meridian data because Meridian has not run them; their retention effects are assumptions and E20 shows how far they can move before the ranking changes. *Source:* analysis/08_cost_worksheet.py

E20. Sensitivity option b

E20. Option (b) stays cheaper than the status quo unless a timed section adds roughly a week of vacancy per hire without buying the full legacy retention gain



Sample: Worksheet evaluated over a grid of r (status quo to pre-AI level) and extra days to fill; other inputs at 2025 averages; quality excluded.

Source: analysis/08_cost_worksheet.py

E20b. Sensitivity table: cost per retained employee under option (b), \$ excl. quality, by r (rows) and extra days to fill (columns)

r	0	5	10	15
0.820	23,377	24,711	26,045	27,379
0.830	23,052	24,370	25,687	27,004
0.840	22,739	24,040	25,341	26,642
0.851	22,430	23,715	25,000	26,285
0.861	22,128	23,398	24,667	25,937
0.871	21,834	23,088	24,343	25,597
0.882	21,546	22,786	24,026	25,265

Sample: 2025-average cost rates. Source: analysis/08_cost_worksheet.py

E20c. Break-even: extra vacancy days that erase option (b)'s saving; recovery share option (c) needs

r assumed	label	breakeven extra days excl. quality	breakeven extra days incl. quality	option c recovery share needed to break even excl. quality	at extra recruiter cost per hire \$	at extra days to fill
0.843	DiD gain only	2.746	9.82	0.890	204.1	3
0.857	legacy 2024-25 observed	4.429	11.62	—	—	—
0.866	legacy gap matched on size and region (nn2, E15f)	5.459	12.72	—	—	—
0.882	pre-AI level	7.38	14.78	—	—	—

Sample: 2025-average rates; status quo from E19 row 0. *Notes:* Each extra day to fill costs vacancy_cost_per_day / r per retained employee. *Source:* analysis/08_cost_worksheet.py